

Oliver's Calculus BC Cram Study Guide

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Chapter 1

Limits

The **limit** of $f(x)$ as x approaches c is, informally, the value that $f(x)$ becomes arbitrarily close to as x approaches the value of c (from both sides). Note that $f(x)$ need not equal the limit at c .

The limit of a function at a particular value may not exist. This may be due to several reasons. It could be that $f(x)$ approaches a different number on the right side as opposed to the left side, it could be that $f(x)$ increases or decreases without bound as x approaches c , or it could be that $f(x)$ oscillates infinitely fast between two fixed values as x approaches c . Learning these common types of reasons that the limit may not exist will aid tremendously in analyzing graphs.

The **squeeze theorem** is a tremendously useful theorem in analyzing the behavior of limits. We present the theorem as follows:

Theorem 1 (Squeeze Theorem). *If $h(x) \leq f(x) \leq g(x)$ for all x in an open interval containing c , except possibly at c itself, and if*

$$\lim_{x \rightarrow c} h(x) = L = \lim_{x \rightarrow c} g(x) \tag{1.1}$$

then $\lim_{x \rightarrow c} f(x)$ exists and is equal to L .

To use the squeeze theorem, you have to be rigorous. You need to show that $h(x) \leq f(x) \leq g(x)$ on an *open interval containing* c , except possibly at c . Then, you can say, “by the squeeze theorem,” and draw the expected conclusion.

The squeeze theorem allows us to find two particularly special trigonometric limits:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad (1.2)$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0 \quad (1.3)$$

We will not go over the proofs for these limits, but just know they exist. Also, you should be able to work with these limits as well. If you can find the following limits, you should be good:

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{x} \quad (1.4)$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{3x} \quad (1.5)$$

Let’s talk about continuity. A function f is **continuous** at a point c if $f(c)$ exists, $\lim_{x \rightarrow c} f(x)$ exists, and $\lim_{x \rightarrow c} f(x) = f(c)$. Furthermore, a function can be continuous on an open interval if it is continuous at every point inside the interval. If it is continuous on $(-\infty, \infty)$, we call the function everywhere continuous.

Functions can also be discontinuous. There are two types of discontinuities: removable and non-removable discontinuities. If you can define or redefine $f(c)$ such that $\lim_{x \rightarrow c} f(x) = f(c)$, then the discontinuity is called **removable**. If not, then the function is non-removable. For example, if there is a simple hole in an otherwise continuous function, then the discontinuity is removable.

We can have limits from the right and limits from the left. A **limit from the right** is the value that $f(x)$ approaches when x approaches c from values

greater than c :

$$\lim_{x \rightarrow c^+} f(x) = L \quad (1.6)$$

Similarly, there exists a **limit from the left**, which is defined appropriately:

$$\lim_{x \rightarrow c^-} f(x) = L \quad (1.7)$$

If the limits from the left and right agree at a given value, then the limit as x approaches that value exists and equals the one-sided limits.

There exists another important theorem that you may want to be aware of, called the intermediate value theorem.

Theorem 2 (Intermediate Value Theorem). *If f is continuous on the closed interval $[a, b]$ and k is any number between $f(a)$ and $f(b)$, then there is at least one number c in $[a, b]$ such that $f(c) = k$.*

This theorem is particularly good at proving the existence of zeros of polynomials in a given interval. If you don't remember what a zero is, go back to algebra class.

Infinite limits are also a thing. A limit in which $f(x)$ increases or decreases without bound as x approaches c is called an **infinite limit**.

Infinite limits are not to be confused with **limits at infinity**, in which one discusses the end behavior of a function (if f approaches a certain value as x increases or decreases without bound). The notation used is

$$\lim_{x \rightarrow \pm\infty} f(x) = L \quad (1.8)$$

where $\pm\infty$ can either be positive or negative infinity.

Chapter 2

Derivatives

The derivative is one of the most important tools of calculus, used in a wide variety of disciplines. Let's review the basic properties of the derivative. The **derivative** of f at x is

$$\frac{d}{dx}f(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \quad (2.1)$$

We say that a function is **differentiable** at a given point if its derivative exists at that point. Furthermore, if a function is differentiable at a point, then it is continuous at that same point.

Memorize these. Constant rule:

$$\frac{d}{dx}(c) = 0 \quad (2.2)$$

Power rule:

$$\frac{d}{dx}(x^n) = nx^{n-1} \quad (2.3)$$

Constant multiple:

$$\frac{d}{dx}(cf(x)) = c \frac{d}{dx}f(x) \quad (2.4)$$

Sum and difference rules:

$$\frac{d}{dx}(f(x) \pm g(x)) = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x) \quad (2.5)$$

Product rule:

$$\frac{d}{dx}(f(x)g(x)) = f(x)\frac{d}{dx}g(x) + g(x)\frac{d}{dx}f(x) \quad (2.6)$$

Quotient rule:

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)\frac{d}{dx}f(x) - f(x)\frac{d}{dx}g(x)}{(g(x))^2} \quad (2.7)$$

When I think of the quotient rule, I think of the following in my head: “low d high minus high d low all over low squared.” It’s kind of stupid but it helps me remember.

Trigonometric functions.

$$\frac{d}{dx}\sin x = \cos x, \quad \frac{d}{dx}\cos x = -\sin x, \quad (2.8)$$

$$\frac{d}{dx}\tan x = \sec^2 x, \quad \frac{d}{dx}\cot x = -\csc^2 x, \quad (2.9)$$

$$\frac{d}{dx}\sec x = \sec x \tan x, \quad \frac{d}{dx}\csc x = -\csc x \cot x \quad (2.10)$$

One can also take **higher-order derivatives**, which are found by taking the derivative repeatedly. For instance, you can find the second derivative, $\frac{d^2}{dx^2}$, by taking the derivative of the derivative.

One of the most important rules in calculus is that of the **chain rule**. We have that if $y = f(u)$ is a differentiable function of u and $u = g(x)$ is a differen-

tiable function of x , then $y = f(g(x))$ is differentiable and

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad (2.11)$$

What happens when we have both the dependent and independent variables in the same expression, but we wish to differentiate? Then, we have to use **implicit differentiation**. The method for conducting implicit differentiation is simple:

1. Differentiate both sides of the equation with respect to x
2. Collect all terms involving dy/dx on the left side of the equation and move all other terms to the right side of the equation
3. Factor dy/dx out of the left side of the equation
4. Solve for dy/dx

Let me illustrate with a simple example: consider the implicit function $x^2 + xy^2 = 1$. Let us find dy/dx . First, we differentiate with respect to x :

$$2x + \left(2xy \frac{dy}{dx} + y^2 \right) = 0 \quad (2.12)$$

Then, we collect the terms:

$$2xy \frac{dy}{dx} = -2x - y^2 \quad (2.13)$$

Finally, we solve:

$$\frac{dy}{dx} = -\frac{1}{y} - \frac{y}{2x} \quad (2.14)$$

Let's talk about **related rates**. These can be tricky. But the method to solve these types of problems can be boiled down into this one strategy.

1. Understand the question. Read it carefully, and I would heavily recommend drawing a picture if applicable to visualize the problem.
2. Write down what information you have (what variables and derivatives you know), and what information you want to know (what variable or derivative you're looking for).
3. Find an appropriate formula describing the situation. This is usually geometric in nature, possibly an area or volume formula, or perhaps a similarity ratio.
4. Conduct implicit differentiation.
5. By this point, if you have followed these steps well, you should be able to use algebra to reach your desired answer.

The derivative of the natural logarithm can be found below:

$$\frac{d}{dx} \ln x = \frac{1}{x}, \quad x > 0 \quad (2.15)$$

To find the derivative of an inverse function, simply use the following formula:

$$\frac{df^{-1}}{dx} = \frac{1}{f'(f^{-1}(x))} \quad (2.16)$$

The derivative of the exponential function is rather simple actually:

$$\frac{d}{dx} e^x = e^x \quad (2.17)$$

Now, for bases other than e for exponential and logarithmic functions, we have:

$$\frac{d}{dx} a^x = (\ln a)a^x, \quad \frac{d}{dx} \log_a x = \frac{1}{(\ln a)x} \quad (2.18)$$

One last thing you may want to know, are inverse trigonometric functions.

The derivatives of such are listed below:

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1-x^2}} \quad (2.19)$$

$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}, \quad \frac{d}{dx} \operatorname{arccot} x = -\frac{1}{1+x^2} \quad (2.20)$$

$$\frac{d}{dx} \operatorname{arcsec} x = \frac{1}{|x|\sqrt{x^2-1}}, \quad \frac{d}{dx} \operatorname{arccsc} x = -\frac{1}{|x|\sqrt{x^2-1}} \quad (2.21)$$

Chapter 3

Applications of Derivatives

Let's talk about one-dimensional physics. A **position function** $s(t)$ is a function that gives the position (in this case, one-dimensional) of some object at a time t . The **velocity** will be found by taking the derivative of the position function (the **speed** is the absolute value of the velocity) and the **acceleration** will be found by taking the derivative of the velocity function.

So hopefully you know what a minimum and maximum are. Let's move onto another important theorem called the extreme value theorem.

Theorem 3 (Extreme Value Theorem). *If f is continuous on a closed interval $[a, b]$, then f has both a minimum and a maximum on the interval.*

Note that continuity on a closed interval means that the one-sided limits of f as x approaches a or b match $f(a)$ or $f(b)$, respectively; additionally, f is continuous at every point between but not including a and b .

Now, a **relative maximum** is a value $f(c)$ at which there exists some open interval containing c such that $f(c)$ is a maximum on that specific interval. A **relative minimum** is defined similarly.

A **critical number** c of a function f is a value at which $\frac{df}{dx}$ at c is equal

to 0 or it does not exist. One will note that if f has a relative minimum or maximum at $x = c$, then c must be a critical number of f .

You will be asked to find extrema (maxima and minima) on a closed interval $[a, b]$. This is not as straightforward. Here's what to do.

1. Find the critical numbers of f in the open interval (a, b)
2. Evaluate f at each critical number
3. Evaluate f at a and b (the endpoints)
4. Which one is the biggest? Which one is the smallest? There you have it: your maximum and your minimum.

Rolle's theorem is another important theorem in calculus. It states the following:

Theorem 4 (Rolle's Theorem). *Let f be continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) . If*

$$f(a) = f(b) \tag{3.1}$$

then there is at least one number c in (a, b) such that $\frac{df}{dx}$ at c is 0.

Additionally, Rolle's theorem can be used to prove the **mean value theorem**, which is defined like so:

Theorem 5 (Mean Value Theorem). *If f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) , then there exists a number c in (a, b) such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a} \tag{3.2}$$

You will be asked about increasing/decreasing functions and concavity. It is important to review the tools at your disposal. Consider the following:

1. If $\frac{df}{dx} > 0$ for all x in (a, b) , then f is increasing on (a, b) .
2. If $\frac{df}{dx} < 0$ for all x in (a, b) , then f is decreasing on (a, b) .
3. If $\frac{df}{dx} = 0$ for all x in (a, b) , then f is constant on (a, b) .

Now let's talk about the **first derivative test**. Suppose c is a critical number of f , that is differentiable on an open interval containing c . Then, if $\frac{df}{dx}$ changes from negative to positive at c , then $f(c)$ is a relative minimum of f . If $\frac{df}{dx}$ changes from positive to negative at c , then $f(c)$ is a relative maximum of f . If $\frac{df}{dx}$ does not change sign at c , then $f(c)$ is neither.

Furthermore, you will be asked about concavity. A function is **concave upward** on some interval if the derivative is increasing on the interval, and **concave downward** if the derivative is decreasing. So, if $\frac{d^2f}{dx^2} > 0$ on the interval, then the graph is concave upward. If $\frac{d^2f}{dx^2} < 0$ on the interval, then the graph is concave downward. A **point of inflection** is a point at which the concavity changes. In that case, either $\frac{d^2f}{dx^2} = 0$ or the second derivative does not exist at that point.

The **second derivative test** is presented as follows. Suppose $f'(c) = 0$ (the derivative of f at c is 0) and the second derivative exists.

1. If $f''(c) > 0$, then $f(c)$ is a relative *minimum*
2. If $f''(c) < 0$, then $f(c)$ is a relative *maximum*

Note that it seems as if it is “flipped,” in a sense. The positive second derivative means a minimum, and the negative second derivative means a maximum. Note that this test has nothing to say when $f''(c) = 0$. You will have to try the first derivative test instead to get a real answer.

You will have to sketch curves on the examination. In order to do that, you will have to do some preparatory work first.

1. Identify the domain and the range of the function.
2. Determine the intercepts, asymptotes, and symmetry of the graph.
3. Determine relative extrema (maxima, minima), and points of inflection through taking derivatives.

Then, you must rely on your drawing skills. Based on that information, locate points of the function of interest and draw the curve.

You may also have to solve basic optimization problems. First, understand the problem; read and visualize. Then, find an equation that models the situation effectively. You may have to differentiate, and then use the various tools you have learned to find the relevant maximum or minimum. By that point, you should have your answer.

I honestly don't remember **Newton's method** being part of the exam, but I will include it for the sake of completeness. Simply use the following formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (3.3)$$

There also exist mathematical objects called **differentials**, defined like so, assuming $y = f(x)$:

$$dy = f'(x)dx \quad (3.4)$$

Finally, you can approximate a function using its tangent line by this formula:

$$y = f(c) + f'(c)(x - c) \quad (3.5)$$

Let's talk about **L'Hopital's Rule**. If you want to take a limit that ends up being an indeterminate form when you attempt to take the limit directly, you have to use L'Hopital's Rule. Its statement is simple:

Theorem 6 (L'Hopital's Rule). *If the limit of $f(x)/g(x)$ as x approaches c*

produces the indeterminate form $0/0$ (or perhaps another indeterminate form such as ∞/∞), then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)} \quad (3.6)$$

L'Hopital's rule is surprisingly good at finding limits with other indeterminate forms, you just have to get a bit creative. For instance, the indeterminate form $0 \cdot \infty$ responds well to being rewritten as a fraction (i.e., $\lim_{x \rightarrow \infty} e^{-x} \sqrt{x} = \lim_{x \rightarrow \infty} \frac{\sqrt{x}}{e^x}$). When you have an indeterminate form of the form 1^∞ or 0^0 , consider taking the natural logarithm of the limit. Finally, when you have the indeterminate form $\infty - \infty$, you likely have two fractions; combine them into one fraction and you may use L'Hopital's rule much more easily.

Chapter 4

Integrals

So far we've covered derivatives, but derivatives are only one of the two most important topics in calculus. Now it's time to cover integrals. We begin by talking about antiderivatives. A function F is an **antiderivative** of f on an interval if $\frac{dF}{dx} = f(x)$ for all x in the interval. Antiderivatives must take the form $F(x) + C$ where C is a constant.

To solve a differential equation of the form

$$\frac{dy}{dx} = f(x) \tag{4.1}$$

We must antidifferentiate, denoted by the symbol $\int$, and so the general solution is

$$y = \int f(x)dx = F(x) + C \tag{4.2}$$

Some important summation formulas are as follows:

$$\sum_{i=1}^n c = cn \tag{4.3}$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} \quad (4.4)$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} \quad (4.5)$$

$$\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4} \quad (4.6)$$

Let us discuss upper and lower sums. These rely on the concept of inscribed and circumscribed rectangles. We have that $f(m_i)$ is the minimum value of $f(x)$ in the i th subinterval, and $f(M_i)$ is the maximum value of $f(x)$ in the i th subinterval. Then, the lower sum is

$$s(n) = \sum_{i=1}^n f(m_i)\Delta x \quad (4.7)$$

and the upper sum is

$$S(n) = \sum_{i=1}^n f(M_i)\Delta x \quad (4.8)$$

A **Riemann sum** is defined as follows: let f be defined on the closed interval $[a, b]$, and let Δ be a partition of $[a, b]$ given by

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b \quad (4.9)$$

where Δx_i is the width of the i th subinterval. If c_i is any point in the i th subinterval, then the sum

$$\sum_{i=1}^n f(c_i)\Delta x_i, \quad x_{i-1} \leq c_i \leq x_i \quad (4.10)$$

is called a Riemann sum of f for the partition Δ .

Riemann sums can be used to define definite integrals. If the limit

$$\lim_{||\Delta|| \rightarrow 0} \sum_{i=1}^n f(c_i) \Delta x_i = \int_a^b f(x) dx \quad (4.11)$$

exists, then f is considered to be **integrable** on $[a, b]$ and the limit is called the **definite integral** of f from a to b . Note that if a function is continuous, then it is integrable.

Furthermore, we have that if f is defined at $x = a$, then

$$\int_a^a f(x) dx = 0 \quad (4.12)$$

Also, if f is integrable on $[a, b]$, then

$$\int_b^a f(x) dx = - \int_a^b f(x) dx \quad (4.13)$$

Let us discuss the **fundamental theorem of calculus**. This is one of the most important theorems in calculus.

Theorem 7 (Fundamental Theorem of Calculus). *If a function f is continuous on the closed interval $[a, b]$ and F is an antiderivative of f on the interval $[a, b]$, then*

$$\int_a^b f(x) dx = F(b) - F(a) \quad (4.14)$$

Furthermore, if f is continuous on an open interval containing a , then, for every x in the interval,

$$\frac{d}{dx} \left(\int_a^x f(t) dt \right) = f(x) \quad (4.15)$$

Note that, in the case of finding the derivative of integrals like

$$F(x) = \int_{\pi/2}^{x^3} \cos t \, dt \quad (4.16)$$

you will have to use the chain rule in addition to the fundamental theorem of calculus.

The **average value** of f on an interval is

$$\frac{1}{b-a} \int_a^b f(x) dx \quad (4.17)$$

Let's discuss u -substitution. If $u = g(x)$, then we have that

$$\int f(g(x))g'(x)dx = \int f(u)du = F(u) + C \quad (4.18)$$

The trickiest part about u -substitution is figuring out when and how to use u -substitution. This comes with practice.

Some tricks that may help you integrate quicker: if f is even, then

$$\int_{-a}^a f(x)dx = 2 \int_0^a f(x)dx \quad (4.19)$$

If f is odd, then

$$\int_{-a}^a f(x)dx = 0 \quad (4.20)$$

Numerical integration is a great way to approximate integrals if you aren't able to integrate regularly. There are two rules that you should know for the exam. The **trapezoid rule** is given by the formula

$$\int_a^b f(x)dx \approx \frac{b-a}{2n} [f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)] \quad (4.21)$$

Meanwhile, we have **Simpson's rule** as well:

$$\int_a^b f(x)dx \approx \frac{b-a}{3n} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 4f(x_{n-1}) + f(x_n)] \quad (4.22)$$

The following is the log rule for integration:

$$\int \frac{1}{x} dx = \ln |x| + C \quad (4.23)$$

We can also find integrals of the six trigonometric functions:

$$\int \sin x dx = -\cos x + C, \quad \int \cos x dx = \sin x + C \quad (4.24)$$

$$\int \tan x dx = -\ln |\cos x| + C, \quad \int \cot x dx = \ln |\sin x| + C \quad (4.25)$$

$$\int \sec x dx = \ln |\sec x + \tan x| + C, \quad \int \csc x dx = -\ln |\csc x + \cot x| + C \quad (4.26)$$

Additionally, the integral of the exponential function is exactly what you would expect it to be:

$$\int e^x dx = e^x + C \quad (4.27)$$

Finally, there are integrals involving inverse trigonometric functions as well:

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin \frac{x}{a} + C \quad (4.28)$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan \frac{x}{a} + C \quad (4.29)$$

$$\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \operatorname{arcsec} \frac{|x|}{a} + C \quad (4.30)$$

Chapter 5

Applications of Integrals

Suppose we want to find the area between two curves. If we have continuous functions f and g on $[a, b]$ and $g(x) \leq f(x)$ everywhere in $[a, b]$, then the area of the region bounded by f and g and the lines $x = a$ and $x = b$ is given by

$$A = \int_a^b [f(x) - g(x)] dx \quad (5.1)$$

To find the volume of a solid of revolution using the **disk method**, we use one of these formulas:

$$V = \pi \int_a^b [R(x)]^2 dx \quad V = \pi \int_c^d [R(y)]^2 dy \quad (5.2)$$

The **washer method** is a variant of the disk method where the solid is bounded by an **outer radius** $R(x)$ and an **inner radius** $r(x)$:

$$V = \pi \int_a^b ([R(x)]^2 - [r(x)]^2) dx \quad (5.3)$$

(Of course, there is an analogous equation for the y -variable.)

If you wish to instead use the **shell method**, use one of the following formulas:

$$V = 2\pi \int_c^d p(y)h(y)dy \quad V = 2\pi \int_a^b p(x)h(x)dx \quad (5.4)$$

where $p(x)$ denotes the radius of the shell, and $h(x)$ denotes the height of the shell.

Distinguishing whether to use the disk or shell method can be tricky to visualize. Think of it this way: if you imagine solid cylinders stacked on top of each other (if you rotate it one way or another), then use the disk method. If you imagine hollow cylinders nested inside each other, then use the shell method. Drawing it out always helps if you are having trouble visualizing how to find the volume in your head.

Suppose that we know that a solid has cross sections of area $A(x)$ or $A(y)$ (you may compute this using geometric formulas). Then, the volume is

$$V = \int_a^b A(x)dx \quad \text{or} \quad V = \int_c^d A(y)dy \quad (5.5)$$

Finally, let's discuss arc length. For a curve $y = f(x)$, the **arc length** is

$$s = \int_a^b \sqrt{1 + [f'(x)]^2}dx \quad (5.6)$$

For a curve $x = g(y)$, the arc length is

$$s = \int_c^d \sqrt{1 + [g'(y)]^2}dy \quad (5.7)$$

Chapter 6

Advanced Integration

To **integrate by parts**, use this formula:

$$\int u dv = uv - \int v du \quad (6.1)$$

Sometimes you will encounter integrals with these expressions: $\sqrt{a^2 - u^2}$, $\sqrt{a^2 + u^2}$, and $\sqrt{u^2 - a^2}$. It will help tremendously to use **trigonometric substitution**; that is, to substitute u with some $\sin \theta$, $\tan \theta$, or $\sec \theta$, and using the formulas

$$\cos^2 \theta = 1 - \sin^2 \theta, \quad \sec^2 \theta = 1 + \tan^2 \theta, \quad \tan^2 \theta = \sec^2 \theta - 1 \quad (6.2)$$

One final technique to solve tricky integrals is to use the method of **partial fractions**. This involves decomposing one large fraction into multiple simpler fractions. For instance, we have:

$$\int \frac{1}{x^2 - 5x + 6} dx = \int \left(\frac{1}{x - 3} - \frac{1}{x - 2} \right) dx = \ln |x - 3| - \ln |x - 2| + C \quad (6.3)$$

In particular, practice will be crucial in order to solve these types of problems

rather quickly. Being able to identify when to use u -substitution, integration by parts, trigonometric substitution, and/or partial fractions is a skill that must be learned through practice. Worst comes to worst, if you come across a multiple-choice question where you cannot integrate, simply take the derivatives of each potential answer until you come across the correct answer.

One particular type of integral that you may come across is the **trigonometric integral**. These are integrals of the form

$$\int \sin^m x \cos^n x dx \quad \int \sec^m x \tan^n x dx \quad (6.4)$$

There are some strategies you may use to solve these equations:

1. If the power of the sine is odd and positive:

$$\begin{aligned} \int \sin^{2k+1} x \cos^n x dx &= \int (\sin^2 x)^k \cos^n x \sin x dx \\ &= \int (1 - \cos^2 x)^k \cos^n x \sin x dx \end{aligned} \quad (6.5)$$

2. If the power of the cosine is odd and positive:

$$\begin{aligned} \int \sin^m x \cos^{2k+1} x dx &= \int \sin^m x (\cos^2 x)^k \cos x dx \\ &= \int \sin^m x (1 - \sin^2 x)^k \cos x dx \end{aligned} \quad (6.6)$$

3. If the powers of both the sine and cosine are even and nonnegative, use the identities

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2} \quad (6.7)$$

to convert the integrand to odd powers of the cosine, then see item 2.

4. If the power of the secant is even and positive:

$$\begin{aligned}\int \sec^{2k} x \tan^n x dx &= \int (\sec^2 x)^{k-1} \tan^n x \sec^2 x dx \\ &= \int (1 + \tan^2 x)^{k-1} \tan^n x \sec^2 x dx\end{aligned}\quad (6.8)$$

5. If the power of the tangent is odd and positive:

$$\begin{aligned}\int \sec^m x \tan^{2k+1} x dx &= \int \sec^{m-1} x (\tan^2 x)^k \sec x \tan x dx \\ &= \int \sec^{m-1} x (\sec^2 x - 1)^k \sec x \tan x dx\end{aligned}\quad (6.9)$$

6. If there are no secant factors and the power of the tangent is even and positive:

$$\begin{aligned}\int \tan^n x dx &= \int \tan^{n-2} x (\tan^2 x) dx \\ &= \int \tan^{n-2} x (\sec^2 x - 1) dx\end{aligned}\quad (6.10)$$

7. For $\int \sec^m x dx$, use integration by parts

8. If all else fails, try converting to sines and cosines

Lastly, let us discuss **improper integrals**. There are several different kinds of improper integrals, all of which will be listed below:

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx \quad (6.11)$$

$$\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx \quad (6.12)$$

$$\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^\infty f(x) dx \quad (6.13)$$

Suppose f has an infinite discontinuity at b . Then

$$\int_a^b f(x)dx = \lim_{c \rightarrow b^-} \int_a^c f(x)dx \quad (6.14)$$

Suppose f has an infinite discontinuity at a . Then

$$\int_a^b f(x)dx = \lim_{c \rightarrow a^+} \int_c^b f(x)dx \quad (6.15)$$

Suppose f has an infinite discontinuity at c in (a, b) . Then

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx \quad (6.16)$$

Chapter 7

Differential Equations

Let's discuss **separation of variables**, which is one very easy trick to solve a differential equation. If the differential equation can be written in the form

$$M(x) + N(y)\frac{dy}{dx} = 0 \quad (7.1)$$

then it is **separable**. Then, what we can do to solve it is arrange it and integrate it like so:

$$\int N(y)dy = \int -M(x)dx \quad (7.2)$$

Regarding **slope fields**, these fields visualize differential equations of the form $y' = F(x, y)$. Note that at any given point (x, y) , one can easily find the slope y' of some solution to the graph. One can easily represent this on the coordinate plane by drawing markings with the slope at each point. To sketch a particular solution, simply just draw the natural curve following the markings of the slope field based on some initial condition.

Euler's method approximates the particular solution of the differential equation $y' = F(x, y)$ that passes through a particular point (x_0, y_0) . Use the

following formula:

$$x_n = x_{n-1} + h \quad y_n = y_{n-1} + hF(x_{n-1}, y_{n-1}) \quad (7.3)$$

Smaller step sizes imply better approximations.

To solve a **first-order linear differential equation**, simply use the following formula:

$$y' + P(x)y = Q(x) \implies ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx}dx + C \quad (7.4)$$

Finally, the **logistic differential equation** is a differential equation of the form

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right) \quad (7.5)$$

where r is the growth rate and K is the carrying capacity. The solution to the logistic differential equation is

$$P(t) = \frac{P_0 K e^{rt}}{(K - P_0) + P_0 e^{rt}} \quad (7.6)$$

where P_0 is initial population.

Chapter 8

Parametric and Polar Equations

Before we move to the calculus of parametric and polar equations, we need to first define what parametric equations are. Let's say we have continuous functions f and g on an interval I . The equations $x = f(t)$ and $y = g(t)$ are called **parametric equations**, whereas t is called the **parameter**.

To find the parametric form of the derivative, if a curve is given by the equations $x = f(t)$ and $y = g(t)$, then the slope of the curve at a given point (x, y) is

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad \frac{dx}{dt} \neq 0 \quad (8.1)$$

Furthermore, we have that for higher derivatives:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{dx/dt} \quad (8.2)$$

$$\frac{d^3y}{dx^3} = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right) = \frac{\frac{d}{dt} \left(\frac{d^2y}{dx^2} \right)}{dx/dt} \quad (8.3)$$

To find arc length in parametric form, this simple formula will prove to be useful:

$$s = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2} dt \quad (8.4)$$

Now let's talk about **polar coordinates**. Polar coordinates rely on two variables, r , the distance from the origin, and θ , the angle in radians (going counterclockwise). Note that $(r, \theta) = (1, 0)$ indicates a distance of 1 to the right. To convert between rectangular and polar coordinates, consider the following formulas:

$$x = r \cos \theta, \quad y = r \sin \theta \quad (8.5)$$

$$\tan \theta = \frac{y}{x}, \quad r^2 = x^2 + y^2 \quad (8.6)$$

To take the derivative of a graph $r = f(\theta)$, you will have to use the above formulas and the chain rule. The end result should look something like this:

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{f(\theta) \cos \theta + f'(\theta) \sin \theta}{-f(\theta) \sin \theta + f'(\theta) \cos \theta} \quad (8.7)$$

Recall that $y = r \sin \theta = f(\theta) \sin \theta$, and $x = f(\theta) \cos \theta$. This should shed light on how the above formula can be derived.

To find area in polar coordinates, for some graph $r = f(\theta)$, use the following formula:

$$A = \frac{1}{2} \int_{\alpha}^{\beta} (f(\theta))^2 d\theta \quad (8.8)$$

Now, to find arc length of a polar curve, we use the following:

$$s = \int_{\alpha}^{\beta} \sqrt{(f(\theta))^2 + (f'(\theta))^2} d\theta = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \quad (8.9)$$

Chapter 9

Series and Sequences

You should already know what sequences are, or else you're probably cooked.

The **limit** of a sequence $\{a_n\}$ is written as

$$\lim_{n \rightarrow \infty} a_n = L \quad (9.1)$$

and if the sequence has a limit, the sequence **converges**. If it does not, the sequence **diverges**.

Do you remember the squeeze theorem? It can also apply for sequences.

Furthermore, it may be useful to note that for a sequence $\{a_n\}$, if

$$\lim_{n \rightarrow \infty} |a_n| = 0 \quad \text{then} \quad \lim_{n \rightarrow \infty} a_n = 0 \quad (9.2)$$

We call a sequence **monotonic** if its terms are nondecreasing or nonincreasing.

A sequence $\{a_n\}$ is considered to be **bounded above** if there is a real number M such that $a_n \leq M$ for all n . M is called an **upper bound** of the sequence.

The least upper bound is known as the **supremum**. Likewise, the definition of **boundedness below** is similar. The definition for **lower bound** is also

similar, and the greatest lower bound is known as the **infimum**.

A **subsequence** of a sequence $\{a_n\}$ is a sequence that is derived from the original sequence simply by deleting some or none of the elements *without* changing the order of the given elements. For instance, an example of a subsequence of $\{a_n\}$ could be $a_1, a_3, a_5, \dots$. The **Bolzano-Weierstrass theorem**, one of the most important theorems in calculus and analysis, states that any bounded sequence (of real numbers) has a convergent subsequence.

If a sequence is bounded and monotonic, then it converges.

Hopefully you also know what a series is. A **partial sum** of an infinite series $\sum a_n$ is given by

$$S_n = a_1 + a_2 + \dots + a_n \quad (9.3)$$

if the sequence of partial sums converges to some number S , the series $\sum a_n$ **converges**. The limit is called the sum of the series. If the sequence of partial sums diverges, the series **diverges**.

You may have learned this in precalculus, but the convergence of a geometric series is given if $0 < |r| < 1$, and

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \quad (9.4)$$

A **telescoping series** is a series of the form

$$(b_1 - b_2) + (b_2 - b_3) + (b_3 - b_4) + (b_4 - b_5) + \dots \quad (9.5)$$

As you might guess, if the series converges, the sum must be

$$S = b_1 - \lim_{n \rightarrow \infty} b_{n+1} \quad (9.6)$$

A series of the form

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \quad (9.7)$$

is called a **p-series**, and it converges if $p > 1$, and diverges if $0 < p \leq 1$.

A series $\sum a_n$ **converges absolutely** if $\sum |a_n|$ converges. If it converges but $\sum |a_n|$ diverges, then it is **conditionally convergent**.

Some important things to know about series:

1. If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.
2. **(nth-Term Test.)** If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges.
3. **(Integral Test.)** If f is positive, continuous, and decreasing for $x \geq 1$ and $a_n = f(n)$, then

$$\sum_{i=1}^{\infty} a_n \quad \text{and} \quad \int_1^{\infty} f(x) dx \quad (9.8)$$

either both converge or both diverge.

4. **(Direct Comparison Test.)** Let $0 < a_n \leq b_n$ for all n . If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges. If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ diverges.
5. **(Limit Comparison Test.)** Suppose that $a_n > 0$, $b_n > 0$, and

$$\lim_{n \rightarrow \infty} \left(\frac{a_n}{b_n} \right) = L \quad (9.9)$$

where L is finite and positive. Then the two series either both converge or both diverge.

6. **(Alternating Series Test.)** Let $a_n > 0$. The alternating series

$$\sum_{n=1}^{\infty} (-1)^n a_n \quad \text{and} \quad \sum_{n=1}^{\infty} (-1)^{n+1} a_n \quad (9.10)$$

converge if the following two conditions are met: $\lim_{n \rightarrow \infty} a_n = 0$ and $a_{n+1} \leq a_n$ for all n .

7. **(Ratio Test.)** Let $\sum a_n$ be a series with nonzero terms. $\sum a_n$ converges absolutely if $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$. $\sum a_n$ diverges if $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1$ or $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \infty$. The test is inconclusive (that is, you have to use another one) if $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$.
8. **(Root Test.)** Let $\sum a_n$ be a series. $\sum a_n$ converges absolutely if we have that $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1$. $\sum a_n$ diverges if $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} > 1$ or $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \infty$. The test is inconclusive if $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = 1$.

To test a series for convergence or divergence, follow these steps:

1. Check if the n th term approaches zero.
2. Check if the series is geometric, p-series, telescoping, or alternating.
3. Check if the integral test, root test, or ratio test can be applied.
4. Check if the series can be compared through the DCT or LCT.

Now let's talk about Taylor and Maclaurin polynomials. If f has n derivatives at c , then the polynomial

$$P_n(x) = f(c) + f'(c)(x - c) + \frac{f''(c)}{2!}(x - c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x - c)^n \quad (9.11)$$

is called the **n th Taylor polynomial** for f at c . If $c = 0$, then

$$P_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots + \frac{f^{(n)}(0)}{n!}x^n \quad (9.12)$$

is called the **n th Maclaurin polynomial** for f . One particularly important theorem in calculus is known as **Taylor's Theorem**:

Theorem 8 (Taylor's Theorem). *If a function f is differentiable through order $n + 1$ in an interval I containing c , then, for each x in I , there exists z between x and c such that*

$$f(x) = f(c) + f'(c)(x - c) + \frac{f''(c)}{2!}(x - c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x - c)^n + R_n(x) \quad (9.13)$$

where

$$R_n(x) = \frac{f^{(n+1)}(z)}{(n+1)!}(x - c)^{n+1} \quad (9.14)$$

The **Lagrange error bound** serves as an estimate for the accuracy of a Taylor or Maclaurin polynomial as opposed to the real function. We have that the Lagrange error bound is:

$$\frac{\max |f^{(n+1)}(z)|}{(n+1)!} |x - c|^{n+1} \quad (9.15)$$

Use your knowledge of functions to compute $\max |f^{(n+1)}(z)|$. The estimate should be reasonable.

A series of the form

$$\sum_{n=0}^{\infty} a_n(x - c)^n \quad (9.16)$$

is called a **power series** centered at c . For any given power series, either:

1. The series converges only at c
2. There exists a **radius of convergence** $R > 0$ such that the series converges absolutely for $|x - c| < R$, and diverges for $|x - c| > R$ (note that it doesn't say anything about $x = c \pm R$), or
3. The series converges absolutely for all x

The **interval of convergence** is not the same as the radius of convergence; this is the set of points where the power series converges. To find the radius

of convergence, often the ratio test is helpful. The radius of convergence of the series obtained by differentiating or integrating a power series is the same as that of the original power series.

Let's talk about **Taylor series**. The series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n \quad (9.17)$$

is called the Taylor series for $f(x)$ at c . If $c = 0$, then it is also called the **Maclaurin series** for f . If we have that $\lim_{n \rightarrow \infty} R_n = 0$ for all x in some interval, then the Taylor series for f converges and equals $f(x)$.

To end this study guide, let's look at some important Maclaurin series:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots \quad (9.18)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + \frac{(-1)^n x^{2n+1}}{(2n+1)!} + \cdots \quad (9.19)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots + \frac{(-1)^n x^{2n}}{(2n)!} + \cdots \quad (9.20)$$

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \cdots, \quad |x| < 1 \quad (9.21)$$